

DLD PROJECT PROPOSAL

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Project Idea A: Main Idea

Project Title: Retro Pong Game

Project Description:

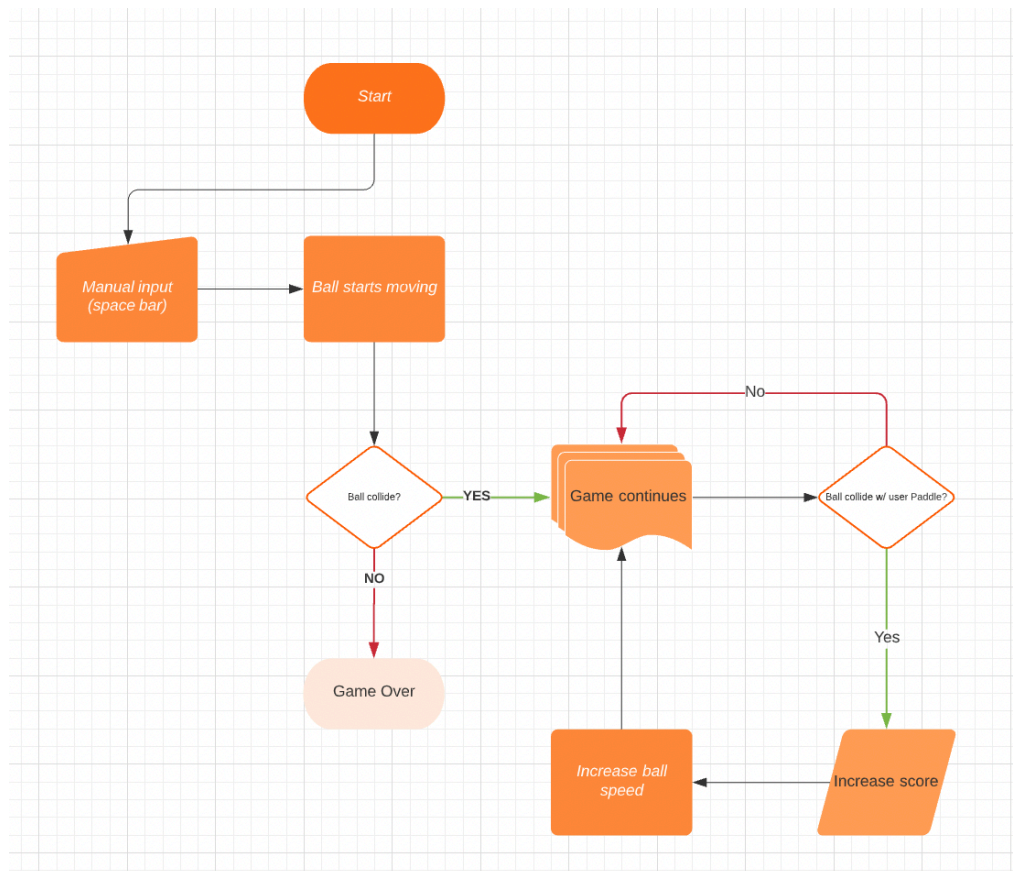
We intend on implementing a simple game of Retro Pong. The game will be digital, projected on a VGA screen with a maximum of one(01) player. It will consist of one paddles and a ball with the objective of preventing the ball from falling below the player's Paddle. If the ball falls below the player's Paddle, the game is over and the total score is displayed.

The player will be able to interact with the Paddle via the Right and Left arrow keys on a keyboard in order to prevent the ball from infiltrating their Safe Zone.

A score will be maintained on the number of times the player is able to prevent the ball from falling through. i.e. Every time the ball collides with the player's Paddle, a number of points will be awarded to them.

Prototype(s)

We will be using Verilog Hardware Development Language (VHDL) alongside a Bass 3 FPGA board to build this game. Below is a high level flow chart of the game's works:



As the game will be displayed on a VGA screen, below is a tentative representation of the UI that it will be using:

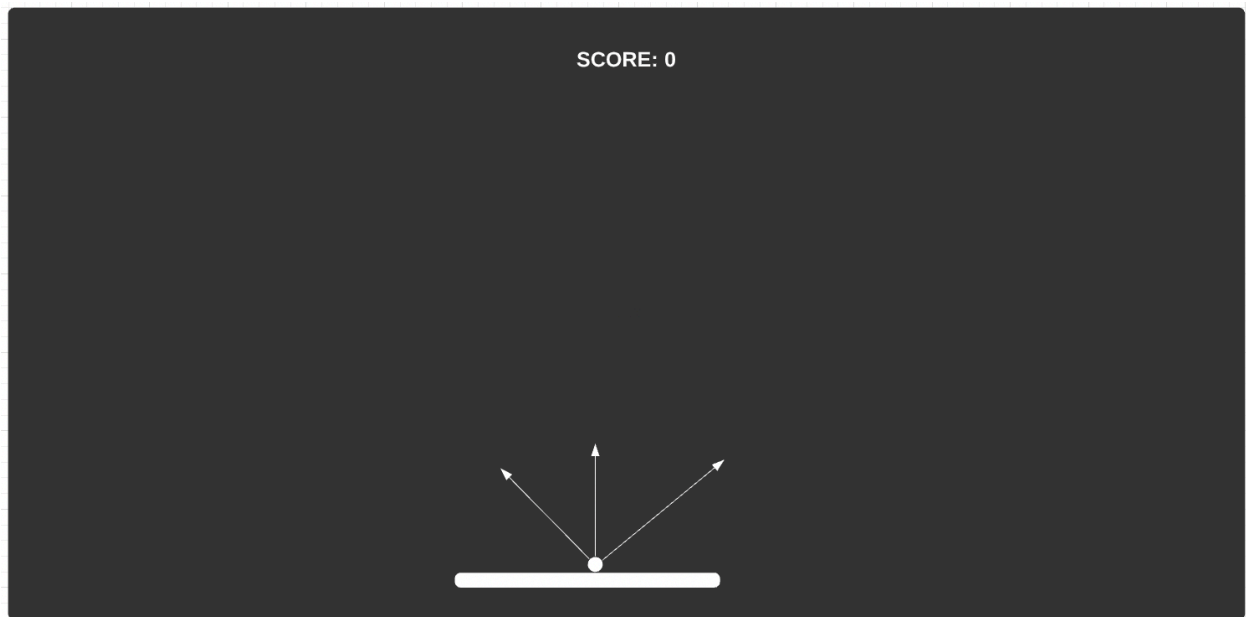


Fig 1

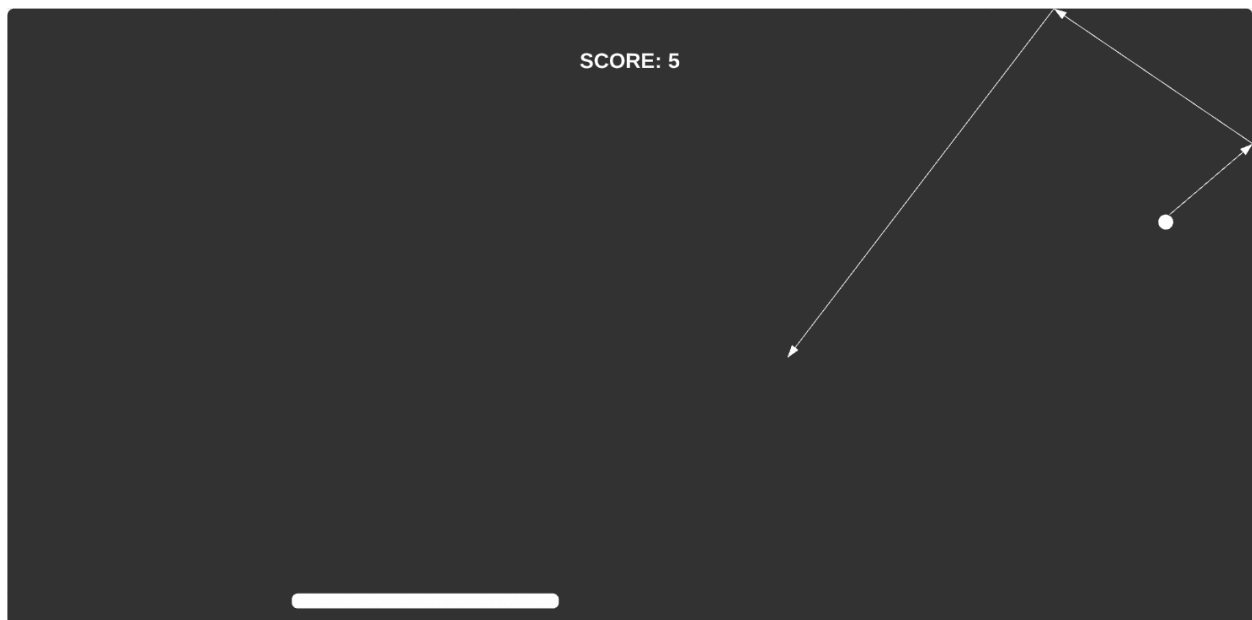


Fig 2

The images above is meant to convey an idea. The actual screens will not display any directional arrows etc.

As can be seen in the images, the ball follows a straight path and may travel in any direction, before colliding into the top, right or left sides of the visible screen. Following this, it will change its trajectory towards the ground. The player will then be required to intercept the ball by moving his Paddle using the Right and Left arrow keys.

Providing input from the keyboard will essentially prompt the system to *re-render* the Paddle on the screen by calculating its new x and y positions based on the player's input. On a successful collision with the Paddle, two high level processes will then execute:

1. The score count will increase
2. The speed of the ball will increase

When the score is increased, the system will once again be prompted to re-render the output to the new value. This will also cause the speed at which the ball moves to increase by a set multiplier, increasing the difficulty of the game and making it a more thrilling experience.

In the event that the player misses the ball, and it falls beneath the puck, it will essentially be colliding with the "Floor". In this case, the System will receive a new prompt, and will kill all running tasks and end the game, updating the screen once more with the final score.

Tentative division of work

We plan on dividing the different modules amongst ourselves equally. Of course, there may be some instances where one of us gets stuck. In such a case, we would rely on peer reviews and peer programming to debug and fix the issue.